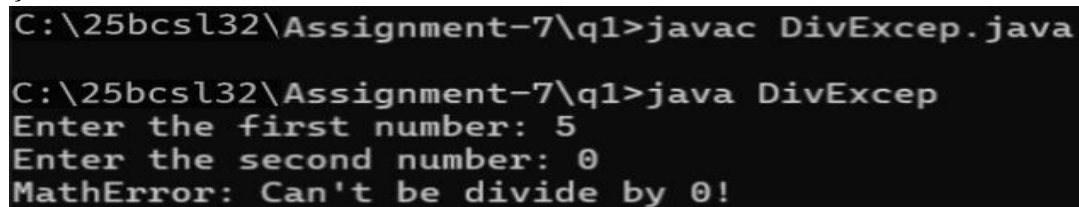


ASSIGNMENT – 7

1. Write an exception handling java program to read two numbers n1, n2 and calculate and print the result of $n1/n2$. If n2 is Zero (0) then it will be handled by exception handler and again ask the value of n2. In the exception handler the program should display appropriate message to the user.

```
import java.util.Scanner;
class DivExcep {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        boolean valid = false;
        System.out.print("Enter the first number: ");
        int n1 = sc.nextInt();
        System.out.print("Enter the second number: ");
        int n2 = sc.nextInt();
        try {
            int res = n1 / n2;
            System.out.println("Result is: "+res);
        }
        catch (ArithmeticException e) {
            System.out.println("MathError: Can't be divide by 0!");
        }
    }
}
```



```
C:\25bcs132\Assignment-7\q1>javac DivExcep.java
C:\25bcs132\Assignment-7\q1>java DivExcep
Enter the first number: 5
Enter the second number: 0
MathError: Can't be divide by 0!
```

2. Write a java program to read two numbers x and y and calculate $x/(x-y)$. The program should check the value of $x-y$. Before dividing with x, it should throw an exception if $x-y$ is zero. In the exception handler the program should display appropriate message to the user.

```
import java.util.Scanner;
class DivCheck {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the first number: ");
        int n1 = sc.nextInt();
        System.out.print("Enter the second number: ");
        int n2 = sc.nextInt();

        try {
```

```

        double res = n1 / (n1-n2);
        System.out.println("Result: "+res);
    }

    catch (ArithmeticException e) {
        System.out.println("Error: "+e+" Can't divide by 0!");
    }
}

```

```

C:\25bcsl32\Assignment-7\q2>javac DivCheck.java
C:\25bcsl32\Assignment-7\q2>java DivCheck
Enter the first number: 5
Enter the second number: 5
Error: java.lang.ArithmeticException: / by zero Can't divide by 0!
C:\25bcsl32\Assignment-7\q2>java DivCheck
Enter the first number: 6
Enter the second number: 5
Result: 6.0

```

3. Write an exception handling java program to print the index position of an existing integer array. The index value will be entered by user. It will be handled by exception handler if index position is greater than the size of array. In the exception handler the program should display appropriate message to the user.

```
import java.util.Scanner;
```

```

public class ArrayIndexExcep {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the array: ");
        int size = sc.nextInt();
        int[] arr = new int[size];

        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println("Array elements are: ");
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }

        try {
            System.out.print("\nEnter the index position to access: ");
            int index = sc.nextInt();
            System.out.println("Element at index " + index + " is: " + arr[index]);
        }
    }
}

```

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```

    }
    catch (ArrayIndexOutOfBoundsException e) {
        System.out.println("Error: Invalid index. Please enter an index
between 0 and " + (arr.length - 1) + ".");
    }
}
}

```

```

C:\25bcsl32\Assignment-7\q3>javac ArrayIndexExcep.java
C:\25bcsl32\Assignment-7\q3>java ArrayIndexExcep
Enter the size of the array: 5
Enter 5 elements:
10 20 30 40 50
Array elements are:
10 20 30 40 50
Enter the index position to access: 6
Error: Invalid index. Please enter an index between 0 and 4.

```

4. Write a program to illustrate the use of multiple catch blocks associated with a single try block.

```
import java.util.Scanner;
```

```

public class MultiCatch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter the first number: ");
            int a = sc.nextInt();
            System.out.print("Enter the second number: ");
            int b = sc.nextInt();

            int result = a / b;
            System.out.println("Result of division: " + result);

            int[] arr = {10, 20, 30};
            System.out.print("Enter an array index to access: ");
            int index = sc.nextInt();
            System.out.println("Element at index " + index + ": " + arr[index]);
        }

        catch (ArithmeticException e) {
            System.out.println("Error: Division by zero is not allowed.");
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Invalid array index.");
        }
    }
}

```

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```

C:\25bcs132\Assignment-7\q4>javac MultiCatch.java
C:\25bcs132\Assignment-7\q4>java MultiCatch
Enter the first number: 5
Enter the second number: 0
Error: Division by zero is not allowed.

C:\25bcs132\Assignment-7\q4>java MultiCatch
Enter the first number: 5
Enter the second number: 2
Result of division: 2
Enter an array index to access: 4
Error: Invalid array index.

```

5. Write a class called Account with the following properties and methods:

Properties: String name, int acc_no, double balance, **Methods:** void deposit (double amt), void withdraw (double amt).

Assume that an account needs to have a minimum balance of 500. If an attempt is made to withdraw, which results in balance going below 500, throw a user defined exception called **MinimumBalanceException**. Use throw and throws wherever necessary.

```
import java.util.Scanner;
```

```

class MinimumBalanceException extends Exception {
    public MinimumBalanceException(String message) {
        super(message);
    }
}

```

```

class Account {
    String name;
    int acc_no;
    double balance;

    Account(String name, int acc_no, double balance) {
        this.name = name;
        this.acc_no = acc_no;
        this.balance = balance;
    }

    void deposit(double amt) {
        balance += amt;
        System.out.println("Amount deposited successfully. Current balance: " + balance);
    }

    void withdraw(double amt) throws MinimumBalanceException {
        if (balance - amt < 500) {
            throw new MinimumBalanceException("Withdrawal denied. Minimum balance of 500 must be maintained.");
        } else {

```

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```

        balance -= amt;
        System.out.println("Amount withdrawn successfully. Current balance: " + balance);
    }
}

```

```

public class BankDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter account holder name: ");
        String name = sc.nextLine();
        System.out.print("Enter account number: ");
        int acc_no = sc.nextInt();
        System.out.print("Enter initial balance: ");
        double balance = sc.nextDouble();

        Account acc = new Account(name, acc_no, balance);

        System.out.print("Enter amount to deposit: ");
        double dep = sc.nextDouble();
        acc.deposit(dep);

        System.out.print("Enter amount to withdraw: ");
        double wd = sc.nextDouble();

        try {
            acc.withdraw(wd);
        } catch (MinimumBalanceException e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}

```

```

C:\25bcs132\Assignment-7\q5>javac BankExcep.java
C:\25bcs132\Assignment-7\q5>java BankExcep
Enter account holder name: Adarsh S. Sahoo
Enter account number: 4632658
Enter initial balance: 5000
Enter amount to deposit: 3000
Amount deposited successfully. Current balance: 8000.0
Enter amount to withdraw: 9000
Error: Withdrawal denied. Minimum balance of 500 must be maintained.

```

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6. Write a program that prompts the user to enter a length in feet and inches and outputs the equivalent length in centimetres. If the user enters a negative number or a non-digit number, throw and handle an appropriate exception and prompt the user to enter another set of numbers.

```
import java.util.Scanner;
```

```
public class MeasureExcep {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int feet = 0, inches = 0;
        boolean valid = false;

        while (!valid) {
            try {
                System.out.print("Enter feet: ");
                String feetInput = sc.next();
                feet = Integer.parseInt(feetInput);
                if (feet < 0) throw new NumberFormatException("Negative number not allowed.");

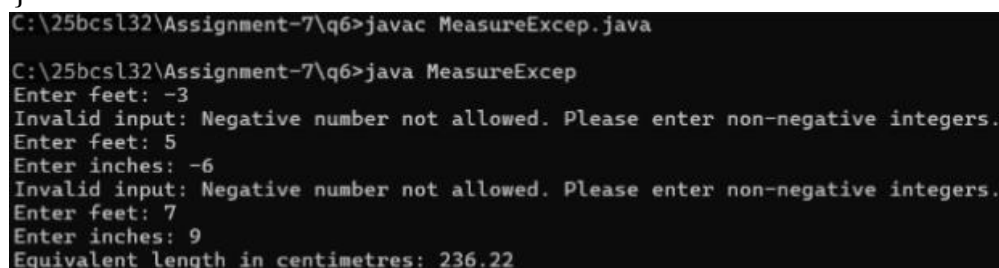
                System.out.print("Enter inches: ");
                String inchInput = sc.next();
                inches = Integer.parseInt(inchInput);
                if (inches < 0) throw new NumberFormatException("Negative number not
allowed.");

                valid = true;

            } catch (NumberFormatException e) {
                System.out.println("Invalid input: " + e.getMessage() + " Please enter non-negative
integers.");
            }
        }

        double totalInches = feet * 12 + inches;
        double cm = totalInches * 2.54;
        System.out.println("Equivalent length in centimetres: " + cm);

        sc.close();
    }
}
```



```
C:\25bcs132\Assignment-7\q6>javac MeasureExcep.java
C:\25bcs132\Assignment-7\q6>java MeasureExcep
Enter feet: -3
Invalid input: Negative number not allowed. Please enter non-negative integers.
Enter feet: 5
Enter inches: -6
Invalid input: Negative number not allowed. Please enter non-negative integers.
Enter feet: 7
Enter inches: 9
Equivalent length in centimetres: 236.22
```

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